

D KARTHIK REDDY

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LinkedIn GitHub

PROFESSIONAL SUMMARY

Engineering student with experience in AI, machine learning, and embedded systems. Worked on deepfake detection, AI-based acoustic monitoring, and UAV surveillance. Developed projects optimizing real-time data processing and achieving high accuracy in classification and detection tasks.

TECHNICAL SKILLS

- Programming Languages: C++, Python, JavaScript
- Web Development: HTML, CSS, Bootstrap, React, Node.js, Express.js
- Machine Learning: TensorFlow, PyTorch, OpenCV, Scikit-learn
- Embedded Systems: Arduino, Raspberry Pi, FPGA, ARM Cortex
- Tools: Git, Docker, Linux, MATLAB, Jupyter Notebook
- Networking: Cisco Packet Tracer, 5G Protocols, Zero-Trust Security

TECHNICAL PROJECTS

AI-based Acoustic Wave Monitoring System

2023

Patent Published

Signal Processing

- Developed an AI-driven acoustic wave monitoring system with 90% accuracy in rail defect detection.
- Processed over 500,000 audio samples to train a deep learning model for real-time fault detection.
- Published a patent for the AI-based monitoring framework, improving predictive maintenance efficiency by 40%.

Deepfake Detection Using SqueezeNet

2023

Research Project

Computer Vision

- Achieved 98.21% validation accuracy using SqueezeNet, trained on 260,000+ images.
- Reduced inference time by 30% through model optimization, enabling real-time analysis of video frames.
- Designed a forensic tool capable of processing 5,000 images in under 14 seconds.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology

2022 – 2026

B.Tech in Electronics and Communication Engineering • CGPA: 9.00/10.0

JNTUH

Minor Degree in Artificial Intelligence and Machine Learning

2023 – Present

VNR VJIE

JNTUH

Sri Chaitanya Junior College

2020 – 2022

Intermediate • Percentage: 98%

Board of Intermediate Education

Sri Chaitanya School

2020

SSC • CGPA: 10.0/10.0

Board of Secondary Education

CERTIFICATIONS & ACHIEVEMENTS

Technical Certifications:

- Neural Networks and Deep Learning - Coursera, Andrew Ng

Technical Achievements:

- Winner of Smart India Hackathon 2023 under the Ministry of Defence, competing among 30,000+ participants.
- Published Patent for AI-based Acoustic Wave Monitoring System, improving defect detection rates by 40%.
- Developed a computer vision library with 100+ GitHub stars and 50+ forks, used by researchers globally.

UPCOMING PROJECTS

LHU-Net for Medical Image Segmentation

Ongoing

Deep Learning

Medical Imaging

- Innovated LHU-Net, leading to a reduction in computational costs by 30 % compared to previous models; this cost-effective solution allowed the lab to reallocate funds toward additional research and development initiatives.
- Processing 10,000+ medical images to optimize segmentation accuracy while maintaining computational efficiency.
- Targeting a 20% reduction in memory usage without sacrificing model performance.